

引导和可控生成

分类

Classifier Guidance

Classifier-Free Guidance

Training-Free Guidance

$$\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t|\mathbf{y}) = \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p_t(\mathbf{y}|\mathbf{x}_t)$$

Score ALD

ILVR

[Conditioning Method for Denoising Diffusion Probabilistic Models](<http://arxiv.org/pdf/2108.02938>)核
心思想是利用一个低通滤波器将参考图像的结构信息作为条件，引导扩散模型的生成过程，而无需重新
训练模型。如果生成的图像与参考图像在低频成分上一致，那么它们在整体结构上就会相似。高频细节
则可以自由生成，实现多样性。

RED-Diff

A VARIATIONAL PERSPECTIVE ON SOLVING INVERSE PROBLEMS WITH DIFFUSION MODELS 暂
时没有看懂

Denoising Diffusion Null Models (DDNM)

Denoising Diffusion Restoration Models (DDRM)

Π GDM

Pseudoinverse-guided diffusion models for inverse problems

假设我们拥有对某个信号 $x_0 \in \mathbb{R}^n$ 的测量值 $y \in \mathbb{R}^m$ ，其关系为

$$y = Hx_0 + z, \quad (2)$$

其中 $H \in \mathbb{R}^{n \times m}$ 是已知的测量矩阵， $z \sim \mathcal{N}(0, \sigma_y^2 I)$ 是一个独立同分布的高斯噪声向量，其各维度上的标准差 σ_y 已知。

我们的目标是求解这个逆问题，从测量值 y 中恢复出原始信号 $x_0 \in \mathbb{R}^n$ 。

$p_t(\mathbf{x}_0|\mathbf{x}_t)$ 使用高斯近似(Π GDM的关键近似):

$$p_t(\mathbf{x}_0|\mathbf{x}_t) \approx \mathcal{N}(\hat{\mathbf{x}}_t, r_t^2 \mathbf{I}), \quad (4)$$

均值使用Tweedie公式估计:

$$\hat{\mathbf{x}}_t = \mathbb{E}[\mathbf{x}_0|\mathbf{x}_t] = \mathbf{x}_t + \sigma_t^2 \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t) \approx \mathbf{x}_t + \sigma_t^2 S_\theta(\mathbf{x}; \sigma_t). \quad (5)$$

其中 $p_t(\mathbf{x}_t|\mathbf{x}_0) \sim \mathcal{N}(\mathbf{x}_0, \sigma_t^2 \mathbf{I})$,注意: r_t 与 σ_t 不同

显然 $p_t(\mathbf{y}|\mathbf{x}_t)$ 也是高斯分布:

$$p_t(\mathbf{y}|\mathbf{x}_t) \approx \mathcal{N}(\mathbf{H}\hat{\mathbf{x}}_t, r_t^2 \mathbf{H}\mathbf{H}^\top + \sigma_y^2 \mathbf{I}).$$

可以推导出 $p_t(\mathbf{y}|\mathbf{x}_t)$ 的得分:

$$\begin{aligned}
\log p_t(\mathbf{y}|\mathbf{x}_t) &= -\frac{1}{2}(\mathbf{y} - H\hat{\mathbf{x}}_t)^\top \Sigma_t^{-1}(\mathbf{y} - H\hat{\mathbf{x}}_t) + \text{const} \quad // \text{取对数, 忽略常数项} \\
\nabla_{\mathbf{x}_t} \log p_t(\mathbf{y}|\mathbf{x}_t) &= -\frac{1}{2} \nabla_{\mathbf{x}_t} ((\mathbf{y} - H\hat{\mathbf{x}}_t)^\top \Sigma_t^{-1}(\mathbf{y} - H\hat{\mathbf{x}}_t)) \quad // \text{记协方差为}\Sigma_t \\
&= -\left(\frac{\partial(\mathbf{y} - H\hat{\mathbf{x}}_t)}{\partial \mathbf{x}_t} \right)^\top \Sigma_t^{-1}(\mathbf{y} - H\hat{\mathbf{x}}_t) \\
&= (H \frac{\partial \hat{\mathbf{x}}_t}{\partial \mathbf{x}_t})^\top \Sigma_t^{-1}(\mathbf{y} - H\hat{\mathbf{x}}_t) \quad // \text{链式法则} \\
&= (H \frac{\partial \hat{\mathbf{x}}_t}{\partial \mathbf{x}_t})^\top (r_t^2 H H^\top + \sigma_y^2 I)^{-1}(\mathbf{y} - H\hat{\mathbf{x}}_t) \quad // \text{带入}\Sigma_t \\
&= \left(\underbrace{(\mathbf{y} - H\hat{\mathbf{x}}_t)^\top (r_t^2 H H^\top + \sigma_y^2 I)^{-1} H}_{\text{vector}} \underbrace{\left(\frac{\partial \hat{\mathbf{x}}_t}{\partial \mathbf{x}_t} \right)}_{\text{Jacobian}} \right)^\top \quad // \Sigma_t \text{是对称矩阵(7)}
\end{aligned}$$

当 $\sigma_y = 0$ 时(无噪声测量),公式7可以近似为:

$$\begin{aligned}
\nabla_{\mathbf{x}_t} \log p_t(\mathbf{y}|\mathbf{x}_t) &\approx \left(\underbrace{(\mathbf{y} - H\hat{\mathbf{x}}_t)^\top (r_t^2 H H^\top)^{-1} H}_{\text{vector}} \underbrace{\left(\frac{\partial \hat{\mathbf{x}}_t}{\partial \mathbf{x}_t} \right)}_{\text{Jacobian}} \right)^\top \\
&= r_t^{-2} \left(\underbrace{(\mathbf{y} - H\hat{\mathbf{x}}_t)^\top (H H^\top)^{-1} H}_{\text{vector}} \underbrace{\left(\frac{\partial \hat{\mathbf{x}}_t}{\partial \mathbf{x}_t} \right)}_{\text{Jacobian}} \right)^\top \\
&= r_t^{-2} \left(\underbrace{(\mathbf{y} - H\hat{\mathbf{x}}_t)^\top (H^\dagger)^\top}_{\text{vector}} \underbrace{\left(\frac{\partial \hat{\mathbf{x}}_t}{\partial \mathbf{x}_t} \right)}_{\text{Jacobian}} \right)^\top \quad // H^\dagger = H^\top (H H^\top)^{-1} \\
&= r_t^{-2} \left((\mathbf{H}^\dagger \mathbf{y} - \mathbf{H}^\dagger \mathbf{H} \hat{\mathbf{x}}_t)^\top \frac{\partial \hat{\mathbf{x}}_t}{\partial \mathbf{x}_t} \right)^\top \quad (8)
\end{aligned}$$

其中: $H^\dagger = H^\top (H H^\top)^{-1}$ 是矩阵H的Moore-Penrose伪逆。

DPS

Diffusion posterior sampling for general noisy inverse problems

CCDF

Come-Closer-Diffuse-Faster: Accelerating Conditional Diffusion Models for Inverse Problems through Stochastic Contraction

TRAINING-FREE LINEAR IMAGE INVERSES VIA FLOWS

TRAINING-FREE LINEAR IMAGE INVERSES VIA FLOWS

将TRAINING-FREE的引导生成从扩散模型引入流匹配模型，并提出一种基于条件最优传输(OT)路径的求解器方案。无需重新训练可适用到现有的扩散模型中。

高斯概率路径:

$$q(\mathbf{x}_t|\mathbf{y}, \mathbf{x}_1) = q(\mathbf{x}_t|\mathbf{x}_1) = \mathcal{N}(\alpha_t \mathbf{x}_1, \sigma_t^2 \mathbf{I}) \quad (5)$$

其中 x_0 为纯噪声, x_1 为数据分布, 对于OT概率路径有: $\alpha_t = 1 - t, \sigma_t = t$ 。

测量值 y 定义如下:

$$y = Ax_1 + \epsilon \quad // \text{其中} \epsilon \sim \mathcal{N}(0, \sigma_y^2 \mathbf{I}) \quad (7)$$

得分引导到速度场引导

由An Introduction to Flow Matching and Diffusion Models的公式54可知速度场和得分存在如下关系

$$u_t(x) = \beta_t^2 \frac{d \ln(\frac{\alpha_t}{\beta_t})}{dt} s_t(x) + \frac{d \ln(\alpha_t)}{dt} x$$

将变量表示替换为本文的有:

$$v(\mathbf{x}_t) = \sigma_t^2 \frac{d \ln(\frac{\alpha_t}{\sigma_t})}{dt} \nabla_{\mathbf{x}_t} \ln q(\mathbf{x}_t) + \frac{d \ln(\alpha_t)}{dt} \mathbf{x}_t$$

响应的条件速度场和条件得分的关系如下:

$$v(\mathbf{x}_t, \mathbf{y}) = \sigma_t^2 \frac{d \ln(\frac{\alpha_t}{\sigma_t})}{dt} \nabla_{\mathbf{x}_t} \ln q(\mathbf{x}_t|\mathbf{y}) + \frac{d \ln(\alpha_t)}{dt} \mathbf{x}_t$$

两式相减有:

$$\begin{aligned}
v(\mathbf{x}_t, \mathbf{y}) - v(\mathbf{x}_t) &= \sigma_t^2 \frac{d \ln(\alpha_t / \sigma_t)}{dt} (\nabla_{\mathbf{x}_t} \ln q(\mathbf{x}_t | \mathbf{y}) - \nabla_{\mathbf{x}_t} \ln q(\mathbf{x}_t)) \\
\Leftrightarrow v(\mathbf{x}_t, \mathbf{y}) - v(\mathbf{x}_t) &= \sigma_t^2 \frac{d \ln(\alpha_t / \sigma_t)}{dt} \nabla_{\mathbf{x}_t} \ln q(\mathbf{y} | \mathbf{x}_t) \quad // \text{加恒为0的} \nabla_{\mathbf{x}_t} \ln q(\mathbf{y}) \\
\Leftrightarrow v(\mathbf{x}_t, \mathbf{y}) &= v(\mathbf{x}_t) + \sigma_t^2 \frac{d \ln(\alpha_t / \sigma_t)}{dt} \nabla_{\mathbf{x}_t} \ln q(\mathbf{y} | \mathbf{x}_t)
\end{aligned} \tag{11}$$

使用PGDM伪逆近似后有:

$$\hat{v}(\mathbf{x}_t, \mathbf{y}) = \hat{v}(\mathbf{x}_t) + \sigma_t^2 \frac{d \ln(\alpha_t / \sigma_t)}{dt} \gamma_t \nabla_{\mathbf{x}_t} \ln q^{app}(\mathbf{y} | \mathbf{x}_t) \tag{12}$$

其中 γ_t 是一个权重因子, 用于调整公式(12)中引导项的强度。

本文假设**先验分布**也是标准高斯分布 $q(x_1) \sim \mathcal{N}(0, I)$, (这是一个很强的假设, 显然先验数据分布不会是高斯分布)。则后验分布的方差为(后面会证明):

$$r_t^2 = \frac{\sigma_t^2}{\sigma_t^2 + \alpha_t^2} \tag{13}$$

后验分布 $q(x_1 | x_t)$ 推导

推导基于一个标准的**贝叶斯推断**过程: 已知**先验分布** $q(x_1)$ 和**似然函数** $q(x_t | x_1)$, 求解**后验分布** $q(x_1 | x_t)$ 的方差参数 r_t^2 。其核心原因是**高斯分布的共轭性质**。

推导步骤与假设

推导基于以下三个关键假设:

1. **数据先验分布**: $q(x_1) = \mathcal{N}(x_1; 0, I)$ 。这是推导的**起点**, 也是得出简洁解析解 $r_t^2 = \frac{\sigma_t^2}{\sigma_t^2 + \alpha_t^2}$ 的原因。
2. **扩散前向过程**: $q(x_t | x_1) = \mathcal{N}(x_t; \alpha_t x_1, \sigma_t^2 I)$ 。这是扩散模型的标准参数化形式, 其中 α_t 和 σ_t 是已知的时间依赖函数。
3. **后验分布形式**: 根据高斯共轭性, 后验 $q(x_1 | x_t)$ 也是高斯分布, 我们将其参数化为 $\mathcal{N}(x_1; \hat{x}_1(x_t), r_t^2 I)$ 。其中 $\hat{x}_1(x_t)$ 是后验均值, r_t^2 是待求的后验方差标量。

下面是详细的推导过程:

目标：求后验分布 $q(x_1|x_t) \propto q(x_t|x_1) \cdot q(x_1)$

1. 写出先验与似然的概率密度函数（忽略常数项）

$$\log q(x_1) = -\frac{1}{2}x_1^\top x_1 + \text{const.} \quad // \text{标准高斯先验}$$

$$\log q(x_t|x_1) = -\frac{1}{2\sigma_t^2}(x_t - \alpha_t x_1)^\top (x_t - \alpha_t x_1) + \text{const.}$$

2. 相加得到非归一化的对数后验

$$\log[q(x_t|x_1)q(x_1)] = -\frac{1}{2} \left[\frac{1}{\sigma_t^2}(x_t - \alpha_t x_1)^\top (x_t - \alpha_t x_1) + x_1^\top x_1 \right] + \text{const}$$

3. 展开并合并关于 x_1 的二次项和一次项

$$\text{展开: } (x_t - \alpha_t x_1)^\top (x_t - \alpha_t x_1) = x_t^\top x_t - 2\alpha_t x_t^\top x_1 + \alpha_t^2 x_1^\top x_1$$

代入并合并：

$$\begin{aligned} \log[\cdot] &= -\frac{1}{2} \left[\frac{\alpha_t^2}{\sigma_t^2} x_1^\top x_1 + x_1^\top x_1 - \frac{2\alpha_t}{\sigma_t^2} x_t^\top x_1 + \frac{1}{\sigma_t^2} x_t^\top x_t \right] + \text{const} \\ &= -\frac{1}{2} \left[\underbrace{\left(\frac{\alpha_t^2}{\sigma_t^2} + 1 \right)}_{\text{二次项系数}} x_1^\top x_1 - \underbrace{\frac{2\alpha_t}{\sigma_t^2} x_t^\top}_{\text{一次项系数}} x_1 \right] + (\text{与 } x_1 \text{ 无关的项}) \end{aligned}$$

4. 与高斯分布的标准形式对比，提取后验方差和均值

$$\begin{aligned} \text{高斯对数密度标准形式: } \log \mathcal{N}(x_1; \mu, \Sigma) &= -\frac{1}{2}(x_1 - \mu)^\top \Sigma^{-1}(x_1 - \mu) + \text{const} \\ &= -\frac{1}{2}x_1^\top \Sigma^{-1}x_1 + x_1^\top \Sigma^{-1}\mu + \text{const.} \end{aligned}$$

对比系数：

$$\text{二次项系数: } \Sigma^{-1} = \left(\frac{\alpha_t^2}{\sigma_t^2} + 1 \right) I = \frac{\sigma_t^2 + \alpha_t^2}{\sigma_t^2} I$$

$$\Rightarrow \Sigma = \frac{\sigma_t^2}{\sigma_t^2 + \alpha_t^2} I$$

$$\text{一次项系数: } \Sigma^{-1}\mu = \frac{\alpha_t}{\sigma_t^2} x_t$$

$$\Rightarrow \mu = \Sigma \cdot \frac{\alpha_t}{\sigma_t^2} x_t = \frac{\sigma_t^2}{\sigma_t^2 + \alpha_t^2} \cdot \frac{\alpha_t}{\sigma_t^2} x_t = \frac{\alpha_t}{\sigma_t^2 + \alpha_t^2} x_t$$

5. 得到结果

$$\text{后验方差: } r_t^2 = \frac{\sigma_t^2}{\sigma_t^2 + \alpha_t^2} \quad // \text{即公式 (13)}$$

$$\text{后验均值: } \hat{x}_1(x_t) = \frac{\alpha_t}{\sigma_t^2 + \alpha_t^2} x_t$$

速度场与干净样本后验估计的关系

已知 $q(x_t|x_1) = \mathcal{N}(x_t; \alpha_t x_1, \sigma_t^2 I)$ 由Tweedies公式有:

$$\begin{aligned} \mathbb{E}[\alpha_t \mathbf{x}_1 | \mathbf{x}_t] &= \mathbf{x}_t + \sigma_t^2 s_\theta(\mathbf{x}; \sigma_t) \\ \Leftrightarrow \alpha_t \mathbb{E}[\mathbf{x}_1 | \mathbf{x}_t] &= \mathbf{x}_t + \sigma_t^2 s_\theta(\mathbf{x}; \sigma_t) \end{aligned}$$

得分与速度场由如下关系:

$$s_t^\theta(x) = \frac{u_t^\theta(x) - \frac{d \ln(\alpha_t)}{dt} x}{\beta_t^2 \frac{d \ln(\frac{\alpha_t}{\sigma_t})}{dt}}$$

带入得分与速度场的关系(并使用本文的表示符号):

$$\begin{aligned} \alpha_t \mathbb{E}[\mathbf{x}_1 | \mathbf{x}_t] &= \mathbf{x}_t + \sigma_t^2 s_\theta(\mathbf{x}; \sigma_t) \\ \Leftrightarrow \alpha_t \hat{x}_1 &= \mathbf{x}_t + \frac{\hat{v} - \frac{d \ln(\alpha_t)}{dt} \mathbf{x}_t}{\frac{d \ln(\frac{\alpha_t}{\sigma_t})}{dt}} \\ \Leftrightarrow \left(\alpha_t \frac{d \ln(\alpha_t / \sigma_t)}{dt} \right) \hat{x}_1 &= \hat{v} + \left(\frac{d \ln(\frac{\alpha_t}{\sigma_t})}{dt} - \frac{d \ln(\alpha_t)}{dt} \right) \mathbf{x}_t \\ \Leftrightarrow \left(\alpha_t \frac{d \ln(\alpha_t / \sigma_t)}{dt} \right) \hat{x}_1 &= \hat{v} - \frac{d \ln \sigma_t}{dt} x_t \\ \Leftrightarrow \hat{v} &= \left(\alpha_t \frac{d \ln(\alpha_t / \sigma_t)}{dt} \right) \hat{x}_1 + \frac{d \ln \sigma_t}{dt} x_t. \end{aligned} \quad (8)$$

不同概率路径的等价性

引理 2. 考虑两条高斯概率路径 q 和 q' , 它们由公式 5 定义, 均值和标准差分别为 α_t, σ_t 和 α'_t, σ'_t 。定义 $t'(t)$ 为给定 t 时, 使得 $\alpha_t / \sigma_t = \alpha'_{t'} / \sigma'_{t'}$ 的唯一解。则:

$$\mathbb{E}_q[\mathbf{x}_1 | \mathbf{x}_t, \mathbf{y}] = \mathbb{E}_{q'}[\mathbf{x}_1 | \mathbf{X}'_{t'(t)} = \alpha'_{t'(t)} \mathbf{x}_t / \alpha_t, \mathbf{y}] \quad (9)$$

这使得我们可以用**任意一种噪声调度的模型** (如预训练的模型) 来模拟另一种噪声调度模型的推理过

程，无需重新训练模型。

证明:

1. 共享先验分布

$$q(x_1|y) = q'(x_1|y)$$

即两个路径在给定条件 y 下，关于 x_1 的先验分布相同。

2. 转移概率的等价性

对于高斯路径，转移概率密度为：

$$\begin{aligned} q(x_t|x_1, y) &= \mathcal{N}(x_t; \alpha_t x_1, \sigma_t^2 I) \\ &\Leftrightarrow x_t = \alpha_t x_1 + \sigma_t z \\ &\Leftrightarrow \frac{x_t}{\alpha_t} = x_1 + \frac{\sigma_t}{\alpha_t} z \end{aligned}$$

类似地:

$$\begin{aligned} q'(x_{t'}|x_1, y) &= \mathcal{N}(x_{t'}; \alpha'_{t'} x_1, \sigma'^2_{t'} I) \\ &\Leftrightarrow x'_{t'} = \alpha'_{t'} x_1 + \sigma'_{t'} z \\ &\Leftrightarrow \frac{x'_{t'}}{\alpha'_{t'}} = x_1 + \frac{\sigma'_{t'}}{\alpha'_{t'}} z \end{aligned}$$

当 $t' = t'(t)$ 满足：

$$\frac{\sigma_t}{\alpha_t} = \frac{\sigma'_{t'}}{\alpha'_{t'}}$$

可以建立对应关系：

$$x'_{t'(t)} = \frac{\alpha'_{t'(t)}}{\alpha_t} x_t$$

两者在给定 x_1, y 时具有相同的分布形式。

因此，在概率密度意义上：

$$q(x_t|x_1, y) = q' \left(x'_{t'(t)} = \frac{\alpha'_{t'(t)}}{\alpha_t} x_t \middle| x_1, y \right)$$

即两个转移概率在重新缩放后一致。

3. 联合分布相同

由于先验 $q(x_1|y) = q'(x_1|y)$ 且转移概率在对应时间点等价（经缩放），因此联合分布满足：

$$q(x_t, x_1|y) = q'(x'_{t'(t)}, x_1|y)$$

在对应变量替换下一致。

4. 条件期望相等

条件期望 $\mathbb{E}[x_1|x_t, y]$ 完全由联合分布 $q(x_1, x_t|y)$ 决定。既然两个路径的联合分布在对应点相同，它们的条件期望也应相同：

$$\mathbb{E}_q[x_1|x_t, y] = \mathbb{E}_{q'}[x_1|x'_{t'(t)}, y]$$

由于信噪比相同，两个路径的线性系数一致，且缩放因子 $\alpha'_{t'(t)}/\alpha_t$ 来自变量对应关系 $x'_{t'(t)} = \frac{\alpha'_{t'(t)}}{\alpha_t} x_t$ 。

算法流程

算法1

预训练的去噪网络,通过条件OT概率路径流获取引导后的样本

Algorithm 1 Solving linear inverse problems via flows using conditional OT probability path

Require: Pretrained denoiser $\widehat{x}_1(x_t)$ converted to conditional OT probability path using Section 3.1, noisy measurement y , measurement matrix A , initial time t , and std σ_y

- 1: Initialize $x_t = ty + (1 - t)\epsilon$, where $\epsilon \sim \mathcal{N}(0, I)$ ▷ Initialize x_t , Eq. (14)
- 2: $z_t = x_t$
- 3: **for** each time step t' of ODE integration **do** ▷ Integrate ODE from $t' = t$ to 1.
- 4: $r_{t'}^2 = \frac{(1-t')^2}{t'^2 + (1-t')^2}$ ▷ Value of r_t^2 from Eq. (13)
- 5: $\widehat{v} = \frac{\widehat{x}_1(z_{t'}) - z_{t'}}{1-t'}$ ▷ Convert $\widehat{x}_1$ to vector field, Eq. (8)
- 6: $g = (y - A\widehat{x}_1)^\top (r_{t'}^2 AA^\top + \sigma_y^2 I)^{-1} A \frac{\partial \widehat{x}_1}{\partial z_{t'}}$ ▷ IIGDM correction
- 7: $\widehat{v}_{\text{corrected}} = \widehat{v} + \frac{1-t'}{t'} g$ ▷ Correct unconditional vector field $\widehat{v}$, Eq. (12)
- 8: **end for**

对于OT路径 $\alpha_t = t, \beta_t = 1 - t$ 速度场:

$$\begin{aligned}
\hat{v} &= \left(\alpha_t \frac{d \ln(\alpha_t/\sigma_t)}{dt} \right) \hat{x}_1 + \frac{d \ln \sigma_t}{dt} x_t \\
&= \left(t \frac{d \ln(t/(1-t))}{dt} \right) \hat{x}_1 + \frac{d \ln(1-t)}{dt} x_t \quad // \alpha_t = t, \beta_t = 1-t \\
&= t \left(\frac{1}{t} - \frac{-1}{1-t} \right) \hat{x}_1 + \frac{-1}{1-t} x_t \\
&= \frac{\hat{x}_1}{1-t} - \frac{x_t}{1-t}
\end{aligned} \tag{8}$$

引导系数:

$$\begin{aligned}
\sigma_t^2 \frac{d \ln(\alpha_t/\sigma_t)}{dt} &= (1-t)^2 \frac{d \ln(t/(1-t))}{dt} \\
&= (1-t)^2 \left(\frac{1}{t} - \frac{-1}{1-t} \right) \\
&= (1-t)^2 \frac{1}{t(1-t)} \\
&= \frac{1-t}{t}
\end{aligned}$$

算法2

预训练的去噪网络,通过等价的概率路径流获取引导后的样本

Algorithm 2 A training-free approach to solve inverse problems via flows with a pretrained denoiser

Require: Pretrained denoiser $\widehat{x}_1(x_t)$ converted to Gaussian probability path with α_t and σ_t using Section 3.1, noisy measurement $\mathbf{y}$, measurement matrix $\mathbf{A}$, initial time t , adaptive weights γ_t , and std σ_y

- 1: Initialize $\mathbf{x}_t = \alpha_t \mathbf{y} + \sigma_t \epsilon$, where $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ ▷ Initialize $\mathbf{x}_t$, Eq. (14)
- 2: $\mathbf{z}_t = \mathbf{x}_t$
- 3: **for** each time step t' of ODE integration **do** ▷ Integrate ODE from $t' = t$ to 1.
- 4: $r_{t'}^2 = \frac{\sigma_{t'}^2}{\sigma_{t'}^2 + \alpha_{t'}^2}$ ▷ Value of r_t^2 from Eq. (13)
- 5: $\widehat{\mathbf{v}} = \left(\alpha_t \frac{d \ln(\alpha_t/\sigma_t)}{dt} \right) \widehat{\mathbf{x}}_1 + \frac{d \ln \sigma_t}{dt} \mathbf{z}_t$ ▷ Convert $\widehat{\mathbf{x}}_1$ to vector field, Eq. (8)
- 6: $\mathbf{g} = (\mathbf{y} - \mathbf{A} \widehat{\mathbf{x}}_1)^\top (r_{t'}^2 \mathbf{A} \mathbf{A}^\top + \sigma_y^2 \mathbf{I})^{-1} \mathbf{A} \frac{\partial \widehat{\mathbf{x}}_1}{\partial \mathbf{z}_{t'}}$ ▷ IIGDM correction
- 7: $\widehat{\mathbf{v}}_{\text{corrected}} = \widehat{\mathbf{v}} + \sigma_t^2 \frac{d \ln(\alpha_t/\sigma_t)}{dt} \gamma_t \mathbf{g}$ ▷ Correct unconditional vector field $\widehat{\mathbf{v}}$, Eq. (12)
- 8: **end for**

算法3

预训练的速度场, 仅增加引导项

Algorithm 3 A training-free approach to solve inverse problems via flows with a pretrained vector field

Require: Pretrained vector field $\widehat{\mathbf{v}}(\mathbf{x}_t)$ converted to Gaussian probability path with α_t and σ_t using Section 3.1, noisy measurement $\mathbf{y}$, measurement matrix $\mathbf{A}$, initial time t , adaptive weights γ_t , and std σ_y

- 1: Initialize $\mathbf{x}_t = \alpha_t \mathbf{y} + \sigma_t \epsilon$, where $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ ▷ Initialize $\mathbf{x}_t$, Eq. (14)
- 2: $\mathbf{z}_t = \mathbf{x}_t$
- 3: **for** each time step t' of ODE integration **do** ▷ Integrate ODE from $t' = t$ to 1.
- 4: $\widehat{\mathbf{v}} = \widehat{\mathbf{v}}(\mathbf{z}_{t'})$ ▷ $\mathbf{z}_{t'}$ is value of $\mathbf{x}_t$ at time t' during ODE integration
- 5: $r_{t'}^2 = \frac{\sigma_{t'}^2}{\sigma_{t'}^2 + \alpha_{t'}^2}$ ▷ Value of r_t^2 from Eq. (13)
- 6: $\widehat{\mathbf{x}}_1 = \left(\alpha_t \frac{d \ln(\alpha_t / \sigma_t)}{dt} \right)^{-1} \left(\widehat{\mathbf{v}} - \frac{d \ln \sigma_t}{dt} \mathbf{z}_t \right)$ ▷ Convert vector field to $\widehat{\mathbf{x}}_1$, Eq. (8)
- 7: $\mathbf{g} = (\mathbf{y} - \mathbf{A} \widehat{\mathbf{x}}_1)^\top (r_{t'}^2 \mathbf{A} \mathbf{A}^\top + \sigma_y^2 \mathbf{I})^{-1} \mathbf{A} \frac{\partial \widehat{\mathbf{x}}_1}{\partial \mathbf{z}_{t'}}$ ▷ Π GDM correction
- 8: $\widehat{\mathbf{v}}_{\text{corrected}} = \widehat{\mathbf{v}} + \sigma_t^2 \frac{d \ln(\alpha_t / \sigma_t)}{dt} \gamma_t \mathbf{g}$ ▷ Correct unconditional vector field $\widehat{\mathbf{v}}$, Eq. (12)
- 9: **end for**

RTC

Real-Time Execution of Action Chunking Flow Policies

$$\mathbf{v}_{\Pi\text{GDM}}(\mathbf{A}_t^\tau, \mathbf{o}_t, \tau) = \mathbf{v}(\mathbf{A}_t^\tau, \mathbf{o}_t, \tau) + \min \left(\beta, \frac{1 - \tau}{\tau \cdot r_\tau^2} \right) \left(\mathbf{Y} - \widehat{\mathbf{A}}_t^1 \right)^\top \text{diag}(\mathbf{W}) \frac{\partial \widehat{\mathbf{A}}_t^1}{\partial \mathbf{A}_t^\tau} \quad (2)$$

其中,

$$\widehat{\mathbf{A}}_t^1 = \mathbf{A}_t^\tau + (1 - \tau) \mathbf{v}(\mathbf{A}_t^\tau, \mathbf{o}_t, \tau), \quad (3)$$

$$r_\tau^2 = \frac{(1 - \tau)^2}{\tau^2 + (1 - \tau)^2}. \quad (4)$$

注意：在RTC中样本值到观测值为恒等映射。 $\mathbf{W}$ 为引导权重, β 是裁剪因子

参考

A Survey on Diffusion Models for Inverse Problems